Distributor Management System

Technologies used and source code structure · Spare-parts distributor (Sri Lanka) · October 2026

1. Overview

A web-based system for a spare-parts distributor. It covers the product catalogue (category → brand → product), stock across warehouses, upcoming shipments, purchasing, sales (quotations, orders, invoices, credit notes), dealer payments with office verification, cheque management, supplier payables, expenses, sales-rep field visits, reports, and Excel/CSV/PDF import and export. Access is controlled by 12 roles with detailed permissions.

It is a single server-rendered web application. Users only need a modern browser (Chrome, Edge, Safari or Firefox). Sales reps use the same site on their phones, so no mobile app has to be installed.

2. Technologies used

Back end (server side)

Technology	Version	Purpose
PHP	8.3+ (built on 8.5)	Programming language
Laravel	13.36	PHP framework: routing, login, permissions, database access, validation
MySQL / MariaDB	MySQL 8+ / MariaDB 10.6+ (dev: MariaDB 10.4)	Relational database (InnoDB, utf8mb4)
Composer	2.x	Installs PHP libraries

Front end (user interface)

Technology	Version	Purpose
Blade	Laravel built-in	Server-side page templates
Tailwind CSS	4	Styling, including green / yellow / red stock colours
Alpine.js	3.14.9	Light interactivity (invoice line items, product search, pop-ups)
Vite + Node.js	Vite 8, Node 20+	Builds CSS and JavaScript (build time only, not needed at runtime)

Libraries for specific features

Library	Version	Used for
openspout/openspout	5.12	Excel (.xlsx) and CSV import / export, shipment uploads
dompdf/dompdf	3.1	PDF invoices, quotations, statements and reports
picqer/php-barcode-generator	3.3	Product barcodes and labels
PHPUnit	Laravel default	Automated tests (45 tests, separate test database)

Laravel features used

Feature	How it is used
Gates & middleware	Role-based permissions for 12 roles; dealer data isolation
Database transactions	Payments, stock and invoices save completely or not at all
Eloquent ORM + migrations	Database models and versioned schema changes
Private file storage	Payment slips, cheque images, product pictures (not publicly reachable)
Database sessions / cache / queue	No Redis required
Mail (SMTP)	Password reset, invoices, dealer notices, overdue reminders, daily summary
Task scheduler	Daily summary email, weekly reminders, housekeeping (needs cron / Task Scheduler)
Audit log	Records who changed what and when

Environments

Environment	Setup
Development	Windows + XAMPP (MariaDB), `php artisan serve`
Production (recommended)	Ubuntu 24.04 LTS, Nginx + PHP-FPM 8.4, MySQL 8 / MariaDB 10.11, HTTPS

3. Source code structure

```
distributor/
├── app/
│   ├── Http/
│   │   ├── Controllers/        34 controllers (one per screen/module)
│   │   └── Middleware/
│   │       └── EnsureRole.php  staff area vs dealer portal
│   ├── Models/                 43 database models
│   ├── Console/Commands/       scheduled tasks: summary, reminders
│   ├── Mail/                   email messages (invoice, payment notice, reset...)
│   ├── Services/               business logic (all money & stock changes)
│   ├── Support/                helpers: permissions, totals, export, settings
│   ├── Providers/
│   └── helpers.php             money(), qty() formatting
├── bootstrap/app.php           middleware aliases
├── config/distributor.php      company rules (limits, thresholds, policies)
├── database/
│   ├── migrations/             18 schema files
│   └── seeders/                DatabaseSeeder (live), DemoDataSeeder (demo only)
├── docs/                       IMPLEMENTATION_PLAN.md, INTEGRATIONS.md
├── public/                     web root (index.php, build/ assets)
├── resources/
│   ├── css/app.css             Tailwind
│   ├── js/app.js               Alpine.js components
│   └── views/                  Blade pages, one folder per module (+ emails/)
├── routes/
│   ├── web.php                 main routes, grouped by permission
│   ├── console.php             schedule of automatic tasks
│   └── modules/                quotations.php, payables.php, expenses.php
├── storage/app/private/        uploaded slips, cheques, product images
├── tests/Feature/              automated tests
├── composer.json / package.json / vite.config.js
└── .env                        server settings (DB, mail, timezone)
```

Services — business logic layer

Service	Responsibility
StockService	The only place stock quantities change; records every movement
SalesService	Sales orders, invoices, credit notes, tax and discounts
PaymentService	Payment recording, duplicate checks, verification workflow, allocation, reversal, cheque deposit / clear / return
PaymentInboxService	Payment slips received from dealers awaiting office review
PurchaseService	Purchase orders and goods receipts
PayablesService	Supplier bills and supplier payments

QuotationService	Quotations and conversion to orders
ShipmentService	Upcoming shipment upload (pending, not in stock) and receiving into stock
ImportService	Excel / CSV imports with all-or-nothing validation
MailService	The only place email is sent; logs every email with its real result (sent / logged only / failed)
DealerNotifier	Emails dealers when a payment is verified / rejected or a cheque is returned (after the database commit)

Support classes

Class	Purpose
Permissions	Permission catalogue and default rights for the 12 roles
Totals	Per-line tax and bill discount calculation
Settings	Company settings stored in the database
Exporter / ExportFormat / SpreadsheetReader	Excel, CSV and PDF export; spreadsheet reading
DocumentNumber	Invoice / order / shipment numbering
Barcode	Barcode generation
ScheduledTasks / DailySummary	The automatic task list and the figures for the morning summary email

Controllers by module

Module	Controllers
Catalogue & stock	ProductController, LookupController (categories, brands), StockController, WarehouseController, ShipmentController
Purchasing	SupplierController, PurchaseOrderController, GoodsReceiptController, SupplierBillController, SupplierPaymentController
Sales	CustomerController, QuotationController, SalesOrderController, InvoiceController, CreditNoteController, DocumentPdfController
Payments & banking	PaymentController, PaymentInboxController, PaymentVerificationController, ChequeController, BankController, CompanyBankAccountController
Sales reps	RepController (rep dashboard, dealer check-in), FieldController (admin live view of reps)
Finance & reports	ExpenseController, ReportController, DashboardController
Data	ImportController (Excel / CSV import), InvoiceEmailController (email invoice PDF)
Administration	UserController, RoleController, SettingsController, AuthController, PasswordResetController, SystemController (scheduled tasks, email log), NoticeController (notice board, holidays)

Models (database tables) by area

Area	Models
Catalogue	Category, Brand, Product, ProductPartNumber, ProductFitment
Stock	Warehouse, StockLevel, StockMovement, Shipment, ShipmentItem
Purchasing	Supplier, PurchaseOrder(+Item), GoodsReceipt(+Item), SupplierBill, SupplierPayment(+Allocation)
Sales	Customer, Quotation(+Item), SalesOrder(+Item), Invoice(+Item), CreditNote(+Item)
Payments	Payment, PaymentAllocation, PaymentSubmission, PaymentStatusHistory, DocumentAttachment
Banking	Bank (30 CBSL-licensed banks), BankBranch, CompanyBankAccount
Expenses	Expense, ExpenseCategory
Users & security	User, Role, Permission, RepVisit, AuditLog, Setting, EmailLog, TaskRun

Views (resources/views)

One folder per module: auth, bank-accounts, banks, cheques, credit-notes, customers, expenses, exports, field, goods-receipts, imports, invoices, lookups, payment-inbox, payment-verification,

payments, pdf (PDF templates), products, purchase-orders, quotations, rep, reports, roles, sales-orders, settings, shipments, stock, supplier-bills, supplier-payments, suppliers, users, warehouses, plus shared `components/` (stock badge, ship badge, layout) and `partials/`.

Automated tests (tests/Feature)

Test file	Covers
AccessControlTest	Role permissions, dealer data isolation
PaymentVerificationTest	Payment workflow, duplicates, segregation of duties, cheques
BankTest	Banks, branches, company accounts
CatalogueAndShipmentTest	Category/brand/product, stock colours, shipments, per-line tax, reps never see quantities
EmailBackupScheduleTest	Password reset, invoice email, dealer notices, mail failures, daily summary, reminders, schedule
SmokeTest	Every main page loads without errors

4. Request flow

```
Browser (PC / phone)
   |  HTTPS
   ▼
Nginx ──► PHP-FPM ──► Laravel  routes/web.php
                         |  middleware: auth, area, can:<permission>
                         ▼
                      Controller  (validates input)
                         ▼
                      Service     (DB transaction, business rules, audit)
                         ▼
                      Models ──► MySQL / MariaDB
                         ▼
                      Blade view  ──► HTML / PDF / Excel back to browser
```

Built: outgoing email (delivers once SMTP details are set in .env), password reset by email, scheduled tasks. Backups are done by the server team (no backup feature in the system).
Not built yet: WhatsApp integration and reading payment slips from an email inbox (need account credentials; see docs/INTEGRATIONS.md).